

Thermodynamics

other words, heat should be taken in at as high a temperature as possible and rejected at as low a temperature as practicable. Hundred per cent efficiency can be achieved only when T_2 reaches absolute zero which is impossible to achieve in practice.

(ii) Eqn. (20.2) also shows that the efficiency of a Carnot engine is independent of the nature of the working substance; it depends only on the temperature of the source and the sink. It also follows from this that *two reversible engines working between the same two temperature limits have the same efficiency.*

(iii) Another important conclusion from the consideration of Carnot's cycle follows from the relation

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2}; \quad \text{or,} \quad \frac{Q_1}{T_1} = \frac{Q_2}{T_2} \quad (20.3)$$

It means that the amount of heat taken in or rejected out are proportional to the absolute temperatures at which they are taken in or rejected out.

(iv) It should be noted that, in practice, it is impossible to construct an engine based on Carnot's cycle which consists of perfectly isothermal and adiabatic operations. This is because, the isothermal process will take place only when the piston will move very slowly so as to allow sufficient time for the transfer of heat to take place. On the other hand, the adiabatic process will take place only when the piston moves very fast so that no heat transfer can take place. It is obvious that such sudden changes in the speed of an engine are not possible in practice.

Beside, in any actual engine, it is impossible to eliminate completely the friction between its various moving parts and loss of heat due to radiation and convection.

Example (20.1) Find the efficiency of a Carnot engine working between the steam point and the ice point.

Soln.

$$\eta = 1 - \frac{T_2}{T_1}$$

for Engineers

$$T_1 = 273 + 100 = 373 \text{ K}$$

$$T_2 = 273 + 0 = 273 \text{ K}$$

$$\begin{aligned} \therefore \eta &= 1 - \frac{273}{373} \\ &= \frac{100}{373} = 0.2681 \end{aligned}$$

$$\% \text{ of efficiency} = 0.2681 \times 100 = \mathbf{26.81\%}.$$

Example 20.2. Find the efficiency of a Carnot engine working between 127°C and 27°C .

Soln.

$$\eta = 1 - \frac{T_2}{T_1}$$

$$T_1 = 273 + 127 = 400 \text{ K}$$

$$T_2 = 273 + 27 = 300 \text{ K}$$

$$\therefore \eta = 1 - \frac{300}{400} = 0.25$$

$$\% \text{ of efficiency} = 0.25 \times 100 = \mathbf{25\%}.$$

Example 20.3. A Carnot engine whose low temperature reservoir is at 7°C has an efficiency of 50%. It is desired to increase the efficiency to 70%. By how many degrees should the temperature of the high temperature reservoir be increased?

Soln.

In the first case,

$$\eta = 50\% = 0.5$$

$$T_2 = 273 + 7 = 280 \text{ K}$$

$$\eta = 1 - \frac{T_2}{T_1}$$

$$\text{or, } 0.5 = 1 - \frac{T_2}{T_1} = 1 - \frac{280}{T_1}$$

$$\text{or, } 0.5 = \frac{280}{T_1}$$

$$\text{or, } T_1 = \frac{280}{0.5} = 560 \text{ K.}$$

In the second case,

$$\eta = 0.7$$

$$T_2 = 280 \text{ K.}$$

$$\eta = 1 - \frac{T_2}{T_1}$$

$$\text{or, } 0.7 = 1 - \frac{280}{T_1} \quad \text{or, } 0.3 = \frac{280}{T_1}$$

$$\text{or, } T_1 = \frac{280}{0.3} = 933.33 \text{ K.}$$

Hence the temperature of the high temperature reservoir should be raised by $(933.33 - 560) = 373.33 \text{ K.}$

Example 20.4. The efficiency of a Carnot engine is $\frac{1}{6}$. If on reducing the temperature of the sink by 65°C , the efficiency becomes $\frac{1}{3}$, find the temperatures of the source and the sink between which the engine is working.

Soln.

Let T_1 and T_2 be the temperatures of the source and sink respectively.

In the first case,

$$\eta = 1 - \frac{T_2}{T_1}; \quad \text{or, } \frac{1}{6} = 1 - \frac{T_2}{T_1} \quad (i)$$

In the second case, temperature of the sink = $T_2 - 65$

$$\eta = 1 - \frac{T_2 - 65}{T_1}$$

$$\text{or, } \frac{1}{3} = 1 - \frac{T_2 - 65}{T_1} \quad (\text{ii})$$

From (i) and (ii) we get

$$\frac{T_2}{T_1} - \frac{T_2 - 65}{T_1} = \frac{1}{3} - \frac{1}{6}$$

$$\text{or, } \frac{65}{T_1} = \frac{1}{6}$$

$$\text{or, } T_1 = 65 \times 6 = 390 \text{ K} = 390 - 273 = 117^\circ\text{C}.$$

From relation (i)

$$1 - \frac{T_2}{390} = \frac{1}{6}$$

$$\text{or, } \frac{T_2}{390} = \frac{5}{6}$$

$$\text{or, } T_2 = \frac{390 \times 5}{6} = 325 \text{ K} = 325 - 273 = 52^\circ\text{C}.$$

Example 20.5 A Carnot engine whose temperature of the source is 400 K takes 200 calories of heat at this temperature and rejects 150 calories of heat in the sink. What is the temperature of the sink? Calculate the efficiency of the engine.

Soln.

From the relation $\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$ we have

$$T_2 = \frac{Q_2}{Q_1} \times T_1$$

$$Q_2 = 200 \text{ cal}$$

$$Q_2 = 150 \text{ cal}$$

$$T_1 = 400 \text{ K.}$$

$$\therefore T_2 = \frac{150}{200} \times 400 = 300 \text{ K.}$$

$$\text{Efficiency } \eta = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{300}{400} = 0.25$$

$$\% \text{ of efficiency} = 0.25 \times 100 = 25\%.$$

Example 20.6 A Carnot engine is operated between two reservoirs at temperatures of 450 K and 350 K. If the engine receives 1 kcal of heat from the source in each cycle, calculate (i) amount of heat rejected to the sink in each cycle, (ii) efficiency of the engine and (iii) work done by the engine in each cycle.

Soln.

$$(i) \quad \frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$

$$\text{or, } Q_2 = \frac{T_2}{T_1} \times Q_1$$

$$T_1 = 450 \text{ K}$$

$$T_2 = 350 \text{ K}$$

$$Q_1 = 1 \text{ kcal} = 1000 \text{ cal.}$$

$$\therefore Q_2 = \frac{350}{450} \times 1000$$

$$= 777.77 \text{ cal.} = 0.778 \text{ kcal.}$$

$$(ii) \quad \eta = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{350}{450} = 0.222\%.$$

$$\% \text{ of efficiency} = 22.2\%.$$

$$\begin{aligned} \text{(iii) Work done} &= Q_1 - Q_2 = 1 - 0.778 \\ &= 0.222 \text{ kcal.} \end{aligned}$$

$$J = 4186 \text{ joules/kcal.}$$

$$\begin{aligned} \therefore W &= 0.222 \times J \text{ joules.} \\ &= 0.222 \times 4186 \text{ joules} \\ &= 930 \text{ joules.} \end{aligned}$$

Example 20.7. An engine works in a Carnot cycle between the temperatures 100°C and 0°C . If the work done in the cycle is 1200 kilogram-metres, find how much heat, measured in calories, is taken in at the higher temperature.

Soln.

If Q_1 is the heat taken in at $(273 + 100) = 373 \text{ K}$, then

$$\frac{Q_1}{Q_2} = \frac{373}{273}; \quad \text{or, } Q_2 = \frac{273}{373} \times Q_1$$

If W is the work done in ergs, then

$W = J \cdot (Q_1 - Q_2)$ where J is the mechanical equivalent of heat.

$$\text{or, } 1200 \times 1000 \times 100 \times 981 = 4.18 \times 10^7 \left(Q_1 - \frac{273}{373} Q_1 \right).$$

$$\begin{aligned} &[1200 \text{ kilogram-metres} = 1200 \times 1000 \times 100 \text{ gm-cm} \\ &= 1200 \times 1000 \times 100 \times 981 \text{ dynes}] \end{aligned}$$

$$\text{or, } 1.2 \times 10^8 \times 981 = 4.18 \times 10^7 \times \frac{100}{373} Q_1$$

$$\begin{aligned} \therefore Q_1 &= \frac{1.2 \times 10^8 \times 981 \times 373}{4.18 \times 10^7 \times 100} \\ &= 1.08 \times 10^4 \text{ calories} \end{aligned}$$

Example (20.8). Find the efficiency of an engine requiring 3×10^6 cal. of heat per horse-power hour and compare it with that of a

perfect reversible engine, assuming that the source is at 1000°C and the sink at 0°C .

Soln.

Amount of work done by the engine in an hour
 $= 746.4 \times 60 \times 60 \text{ joules}$

Efficiency, $\eta = \frac{\text{work done}}{\text{heat supplied}}$

$$= \frac{W \text{ joules}}{Q \text{ cal}} = \frac{W \text{ joules}}{J \cdot Q \text{ joules}}$$

$$= \frac{746.4 \times 60 \times 60}{4.2 \times 3 \times 10^6} = 0.213$$

$$\% \text{ efficiency} = 0.213 \times 100 = 21.3\%$$

Efficiency of the perfect reversible engine

$$= 1 - \frac{T_2}{T_1} = 1 - \frac{(273 + 0)}{(273 + 1000)}$$

$$= 1 - \frac{273}{1273} = 0.785$$

$$\% \text{ efficiency} = 0.785 \times 100 = 78.5\%$$

The ratio of the efficiencies = **21 : 78 (approx.)**

20.10 Refrigerator

A refrigeration may be regarded as a heat engine operating in reverse. Carnot's cycle is perfectly reversible; therefore, it can work as a heat engine and also as a refrigerator. When it works as a heat engine it absorbs an amount of heat Q_1 from the source at a higher temperature T_1 , does an amount of work W and rejects an amount of heat Q_2 to a sink at a lower temperature T_2 . When it works as a refrigerator, it absorbs an amount of heat Q_2 from the sink at a lower temperature T_2 , does an amount of work W and rejects an amount of heat Q_1 to the source at a higher temperature T_1 .

Waves

onward propagation of the wave through the medium.

(ii) The maximum displacement of all particles of the medium is same which they attain one after another.

(iii) All particles pass through their mean positions with the same maximum velocity but one after another.

(iv) No particle of the medium is permanently at rest.

(v) Every region in the path of the wave becomes successively a region of compression, normal pressure and rarefaction. Thus all particles of the medium undergo the same changes of pressure but one after the other.

(vi) A regular transfer of energy takes place across every section of the medium.

remains confined within the space where it is produced.

(ii) The maximum displacement of the particles are not same but they attain their respective maximum displacements simultaneously. The displacement decreases progressively from its maximum value at an antinode to zero at the adjoining node.

(iii) All particles pass through their mean positions simultaneously but with different maximum velocities.

(iv) Certain particles of the medium (the displacement nodes) are permanently at rest.

(v) Condensation and rarefaction do not move along as in the progressive wave. They simply appear and disappear again to be succeeded by the opposite conditions in the same place. The change of pressure is maximum at the nodal points and zero at the antinodal points but occur simultaneously at all points.

(vi) There is no transference of energy across any section of the medium.

Example 5.6. A plane progressive harmonic wave is travelling with a velocity of 340 m/sec in a fluid medium of density 0.0015 gm/sec. If the amplitude of the wave be 10^{-4} cm and its frequency 300 cps, obtain the values of (i) pressure amplitude, (ii) the energy density and (iii) energy current for it.

Soln.

(i) In the case of a plane harmonic wave

$$y = a \sin \frac{2\pi}{\lambda} (vt - x)$$

$$\text{and } p = -K \frac{dy}{dx} \text{ (see Art. 5.7)}$$

$$= K \frac{2\pi}{\lambda} a \cos \frac{2\pi}{\lambda} (vt - x)$$

Obviously p will be maximum when $\cos \frac{2\pi}{\lambda} (vt - x)$ is maximum i.e., is equal to 1.

So the maximum value of p or the pressure amplitude $= K \frac{2\pi}{\lambda} a$.

Now $v = \sqrt{\frac{K}{\rho}}$; or, $K = \rho v^2$ where ρ is the density of the medium.

Hence, pressure amplitude $= \rho v^2 \cdot \frac{2\pi}{\lambda} a$.

Again, from $v = n\lambda$, we have $\frac{1}{\lambda} = \frac{n}{v}$.

$$\therefore \text{ pressure amplitude} = 2\pi \rho v^2 a \cdot \frac{n}{v}$$

$$= 2\pi \rho v a n$$

Here $\rho = 0.0015 \text{ gm/cc}$, $v = 340 \text{ m/sec} = 340 \times 10^2 \text{ cm/sec}$, $a = 10^{-4} \text{ cm}$ and $n = 300 \text{ cps}$.

$$\therefore \text{ pressure amplitude} = 2\pi \times 0.0015 \times 340 \times 10^{-4} \times 300 \\ = 9.6084 \text{ dyne/cm}^2.$$

(ii) Energy density, $E = 2\pi^2 n^2 a^2 \rho$ (Art. 4.9)

$$= 2\pi^2 \times (300)^2 \times (10^{-4})^2 \times 0.0015$$

$$= 2.665 \times 10^{-5} \text{ erg/cc.}$$

(iii) Energy current (or energy flux),

$$I = E \cdot v$$

$$= 2.665 \times 10^{-5} \times 340 \times 10^2$$

$$= 0.9059 \text{ erg/cm}^2 - \text{sec.}$$

Example 5.7 (a). A sound wave in air, having an amplitude of 0.005 cm and frequency 700 cps. travelling along the direction of positive x -axis with a velocity of 350 m/sec suffers reflection at a free boundary. Obtain the values of (i) the displacement amplitude and (ii) the pressure amplitude in the resulting stationary wave at a point $x = 50$ cm. (Density of air = 1.293 gm/litre).

(b) Also obtain the distance between (i) two successive displacement nodes and antinodes, (ii) two successive pressure nodes and antinodes, (iii) a displacement node and an adjacent pressure node.

Soln.

The equation of a stationary wave formed due to reflection at a free boundary is given by

$$y = 2a \cos \frac{2\pi x}{\lambda} \sin \frac{2\pi vt}{\lambda}$$

where the displacement amplitude

$$= 2a \cos \frac{2\pi x}{\lambda}$$

$a = 0.005$ cm, $x = 50$ cm and

$$\lambda = \frac{v}{n} = \frac{350 \times 100}{700} = 50 \text{ cm}$$

$\therefore$ displacement amplitude

$$= 2 \times 0.005 \cos 2\pi \times \frac{50}{50} = 0.01 \text{ cm.}$$

Since the cosine term is equal to 1, the displacement amplitude is maximum and the point is, therefore, a displacement *antinode*.

(ii) excess pressure,

$$p = \rho v^2 \frac{4\pi a}{\lambda} \sin \frac{2\pi x}{\lambda} \sin \frac{2\pi vt}{\lambda}$$

$\therefore$ pressure amplitude

$$= \rho v^2 \frac{4\pi a}{\lambda} \sin \frac{4\pi x}{\lambda}$$

$$= \frac{0.001293 \times (35000)^2 \times 4\pi \times 0.005}{50} \times \sin \frac{2\pi \times 50}{50} = 0$$

Thus the point is a pressure node which is also obvious since the point is a displacement antinode.

(b) Since the distance between two successive displacement nodes or antinodes is $\lambda/2$, the distance between a displacement node and an adjacent displacement antinode is $\lambda/4$ and also because a displacement node coincides with a pressure antinode and vice versa.

distance between two successive displacement nodes = distance between two successive displacement antinodes

$$= \lambda/2 = 50/2 = 25.0 \text{ cm.}$$

Therefore, the distance between two successive pressure nodes (or antinodes) is also $= \lambda/2 = 50/2 = 25.0 \text{ cm.}$

The distance between a displacement node (or pressure antinode) and an adjacent pressure node (or displacement antinode) $= \lambda/4 = 50/2 = 12.5 \text{ cm.}$

EXERCISES

- [1] What is interference of sound waves? Distinguish between constructive and destructive interference.
- [2] Analytically discuss the interference of sound and obtain the condition for maximum and minimum intensity.
- [3] Show that in interference of sound waves, energy is not destroyed – it is simply redistributed from the points of minimum intensity to those of maximum intensity.
- [4] What are beats? How are they produced? Analytically discuss the formation of beats and show that the number of beats per second is equal to the difference in frequency of the two notes.
- [5] Distinguish between phase velocity and group velocity of a train of waves and establish a relation between them.

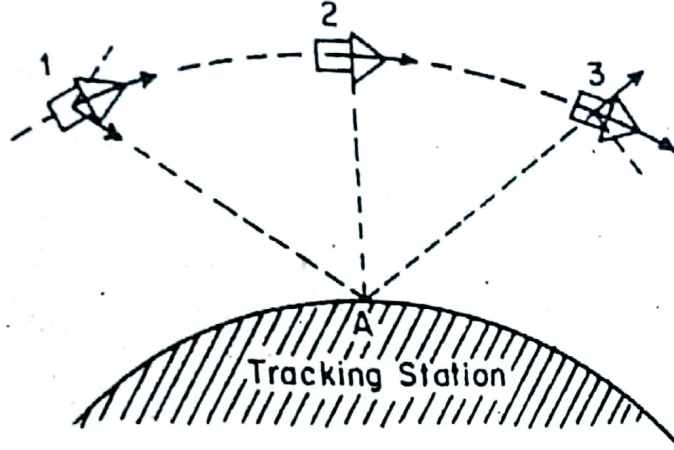


Fig. 7.5

The phenomenon of Doppler effect is used by the traffic police to detect the speed of automobiles. The technique is to emit an electro-magnetic wave of constant frequency by a source attached to the police van. This wave is reflected by the moving car, which in turn acts as a moving source. This results in a Doppler shift in the frequency of the reflected wave. From the shift in frequency, the speed of the car can be measured.

Example (7.1) A train approaches a stationary observer at a speed of 75 kilometers per hour sounding a whistle of frequency 1000. What will be the apparent frequency of the whistle to the observer? (velocity of sound = 330 m/sec)

Soln.

$$n' = \left(\frac{V - v_0}{V - v_s} \right) n$$

Here $n = 1000$, $V = 330 \text{ m. sec}^{-1}$

$$v_0 = 0$$

$$v_s = 75 \text{ km/hr} = 20.833 \text{ m. sec}^{-1}$$

$$\begin{aligned} \therefore n' &= \left(\frac{330}{330 - 20.833} \right) 1000 \\ &= \frac{330}{309.167} \times 1000 = 1067. \end{aligned}$$

Example 7.2 Show that if the source moves away with the velocity of sound from an observer who is at rest, the frequency of vibration is halved.

Soln.

$$n' = \left(\frac{V - v_0}{V - v_s} \right) n$$

here $v_0 = 0$

and $v_s = -V$.

$$\begin{aligned} n' &= \frac{V}{V + V} \cdot n \\ &= \frac{V}{2V} \cdot n = \frac{1}{2} n. \end{aligned}$$

Example 7.3 A car sounding a horn producing a note of 500 Hz, approaches and then passes a stationary observer at a steady speed of 20 m. sec⁻¹. What will be the frequencies apparent to the observer when the car is (i) approaching and (ii) receding? What is the interval between these two notes? (velocity of sound = 340 m. s⁻¹).

Soln.

$$(i) \quad n' = \left(\frac{V - v_0}{V - v_s} \right) n$$

here $V = 340 \text{ m.s}^{-1}$

$v_0 = 0, v_s = 20 \text{ m.s}^{-1}$

$n = 500 \text{ Hz.}$

$$\begin{aligned} \therefore n' &= \frac{V}{V - v_s} \cdot n \\ &= \left(\frac{340}{340 - 20} \right) \times 500 \end{aligned}$$

$$= \frac{340}{320} \times 500 = 531 \text{ Hz.}$$

(ii) here $v_s = -20 \text{ ms}^{-1}$ and $v_o = 0$.

$$\therefore n' = \left[\frac{340}{340 - (-20)} \right] \times 500$$
$$= \frac{340}{360} \times 500 = 472 \text{ Hz.}$$

Interval between the notes

= ratio of the two frequencies

$$= \frac{531}{472} = 1.125$$

Example (7.4) Calculate the velocity at which a source of frequency 10000 per second should approach the observer at rest in order to produce Doppler's shift of 200 per second. (velocity of sound = 340 m per sec.)

Soln.

$$n' = \left(\frac{V - v_o}{V - v_s} \right) n$$

Here $V = 340 \text{ m/sec}$

$$v_o = 0, n = 10,000$$

$$v_s = ?$$

Since the source approaches the observer, v_s is positive and, therefore, $n' > n$.

$$\therefore n' = n + 200 = 10200$$

$$\therefore 10200 = \left(\frac{340}{340 - v_s} \right) \times 10000$$

$$\text{or, } 340 - v_s = \frac{340}{10200} \times 10000$$
$$= 333.33$$

$$\therefore v_s = 340 - 333.33$$
$$= 6.66 \text{ m per sec.}$$

Example 7.5. Calculate the velocity at which an observer should approach a source of frequency one thousand per second at rest in order to produce a Doppler shift of 40 per second. (velocity of sound in air = 340 m/sec).

Soln.

$$n' = \left(\frac{V - v_0}{V - v_s} \right) n$$

Here $n = 1000$,

$v_s = 0$, $V = 340$ m/sec.

Observer moves towards the source i.e., v_0 is negative. Therefore $n' > n$;

or, $n' = 1000 + 40 = 1040$.

$$\begin{aligned} \therefore n' &= \left[\frac{V - (-v_0)}{V} \right] \times n \\ &= \left(\frac{V + v_0}{V} \right) n \end{aligned}$$

$$\begin{aligned} \text{or, } 1040 &= \left(\frac{340 + v_0}{340} \right) \times 1000 \\ &= \left(1 + \frac{v_0}{340} \right) \times 1000 \\ &= 1000 + \frac{1000 v_0}{340} \end{aligned}$$

$$\text{or, } 1040 - 1000 = \frac{1000 v_0}{340}$$

$$\text{or, } \frac{1000 v_0}{340} = 40$$

$$\therefore v_0 = \frac{13600}{1000} = 13.6 \text{ m/sec.}$$

Example 7.6 Two aeroplanes pass each other in opposite directions and one of them is blowing a whistle of frequency 540 Hz.

for English

Calculate the frequencies of the notes heard in the other aeroplane (i) before and (ii) after they have passed each other. Velocity of either of the aeroplanes is 540 km hr^{-1} and the velocity of sound = 350 ms^{-1} .

Soln.

$$V = 350 \text{ m/s.}$$

$$v_s = v_0 = \pm 540 \text{ km/hr} = 150 \text{ m/sec.}$$

$$n = 540 \text{ Hz.}$$

$$\text{and } n' = \left(\frac{V - v_0}{V - v_s} \right) \times n$$

$$(i) \quad v_0 = -ve \text{ and } v_s = +ve.$$

$$\therefore n' = \left(\frac{V + v_0}{V - v_s} \right) \times n$$

$$= \left(\frac{350 + 150}{350 - 150} \right) \times 540$$

$$= \frac{500}{200} \times 540$$

$$= 1350 \text{ Hz.}$$

$$(ii) \quad v_0 = +ve \text{ and } v_s = -ve$$

$$\therefore n' = \left(\frac{V - v_0}{V + v_s} \right) \times n$$

$$= \left(\frac{350 - 150}{350 + 150} \right) \times 540$$

$$= \frac{200}{500} \times 540 = 216 \text{ Hz.}$$

Example 7.7. Two aeroplanes A and B are approaching each other and their velocities are 108 km/hr and 144 km/hr respectively. The frequency of a note emitted by A as heard by the passengers in B is 1170 Hz . Calculate the frequency of the note as heard by the passengers in A (velocity of sound = 350 m/sec).

Soln.

Here $V = 350$ m/sec.

$$v_0 = 144 \text{ km/hr} = 40 \text{ m/sec}$$

$$v_s = 108 \text{ km/hr} = 30 \text{ m/sec}$$

$$n' = 1170 \text{ Hz.}$$

$$n = ?$$

The source and the observer are approaching each other. So v_s is positive and v_0 is negative.

Hence, from $n' = \left(\frac{V - v_0}{V - v_s} \right) \times n$ we have

$$\begin{aligned} n' &= \left(\frac{V + v_0}{V - v_s} \right) \times n \\ &= \left(\frac{350 + 40}{350 - 30} \right) \times n \end{aligned}$$

$$\text{or, } 1170 = \frac{390}{320} \times n$$

$$\begin{aligned} \text{or, } n &= \frac{320 \times 1170}{390} \\ &= 960 \text{ Hz.} \end{aligned}$$

Example (7.8) The apparent frequency of the whistle of an engine changes in the ratio 6 : 5 as the engine passes an stationary observer. If the velocity of sound is 352 m/sec, calculate the velocity of the engine.

Soln.

Let the frequencies as heard by the observer when the train is moving towards and away from him be n_1 and n_2 respectively. Then

$$\frac{n_1}{n_2} = \frac{6}{5}$$

$$\text{Now } n_1 = \left(\frac{V - v_0}{V - v_s} \right) \times n$$

$$= \left(\frac{V}{V - v_s} \right) \times n \quad (\because v_0 = 0 \text{ and } v_s = +ve)$$

$$\text{and } n_2 = \left(\frac{V - v_0}{V - v_s} \right) \times n$$

$$= \left(\frac{V}{V + v_s} \right) \times n \quad (\because v_0 = 0 \text{ and } v_s = -ve)$$

$$\therefore \frac{n_1}{n_2} = \left(\frac{V}{V - v_s} \right) \times \left(\frac{V + v_s}{V} \right)$$

$$= \frac{V + v_s}{V - v_s}$$

$$\text{or, } \frac{6}{5} = \frac{352 + v_s}{352 - v_s}$$

$$\text{or, } v_s = 13.8 \text{ m.s}^{-1}$$

The speed of the engine is 13.8 m/sec.

Example (7.9) A car emitting sound of frequency 200 Hz is moving away from a stationary observer and towards a rigid flat wall. The velocity of the car is 5 m/sec. Calculate the number of beats heard per second by the observer. Velocity of sound in air = 350 m/sec.

Soln.

$$n' = \left(\frac{V - v_0}{V - v_s} \right) n$$

$$n = 200 \text{ Hz.}$$

$$v_0 = 0, v_s = 5 \text{ m/sec.}$$

$$V = 350 \text{ m/sec.}$$

As the source moves away from the observer, v_s is negative.

$$\therefore n_1 = \left(\frac{350}{350 - (-5)} \right) \times 200$$

$$= \frac{350}{355} \times 200 = 197.2 \text{ Hz.}$$

The observer also hears a sound of apparent frequency n_2 due to reflection of sound waves by the wall. This means the source is now moving towards the stationary observer. Or, v_s is positive.

$$\therefore n_2 = \left(\frac{V}{V - v_s} \right) \times n$$

$$= \frac{350}{350 - 5} \times 200$$

$$= \frac{350}{345} \times 200 = 202.9 \text{ Hz.}$$

Therefore, the number of beats produced per second.

$$= n_2 - n_1 = 202.9 - 197.2$$

$$= 5.7.$$

Example (7.10.) A motor car fitted with two sounding horns which differ in frequency by 320 vibrations per second is speeding at the rate of 36 km/hr towards a stationary observer. Calculate the difference in the frequencies of the notes heard by him. Velocity of sound in air = 330 m/sec.

Soln.

Let n_1 and n_2 be the frequencies.

For n_1 , the apparent frequency is

$$n'_1 = \frac{V}{V - v_s} \times n_1 \quad (i)$$

and for n_2 , the apparent frequency is

$$n'_2 = \frac{V}{V - v_s} \times n_2 \quad (ii)$$

Subtracting (ii) from (i), we have

$$n'_1 - n'_2 = \frac{V}{V - v_s} (n_1 - n_2)$$

Here $V = 330$ m/sec.

$$v_s = 36 \text{ km/hr} = 10 \text{ m/sec}$$

$$n_1 - n_2 = 320$$

$$\therefore n'_1 - n'_2 = \frac{330}{330 - 10} \times 320$$

$$= \frac{330}{320} \times 320 = 330 \text{ vibrations per second.}$$

Example 7.11 At each of the two stations A and B, a siren is sounding with a constant frequency of 250 cycles per second. A cyclist from A proceeds straight towards B with a velocity of 12 km/hr and hears 5 beats per second. Calculate the velocity of sound.

Soln.

(i) In the first case the cyclist is moving away from the source A. So v_0 is positive. We, therefore, have

$$n_1 = \left(\frac{V - v_0}{V} \right) n$$

$$v_0 = 12 \text{ km/hr} = \frac{10}{3} \text{ m/sec.}$$

$$n = 250$$

$$\therefore n_1 = \left(\frac{V - \frac{10}{3}}{V} \right) \times 250$$

(ii) In the second case the cyclist is moving towards the source B. So v_0 is negative. Therefore, we have,

$$n_2 = \left(\frac{V - (-v_0)}{V} \right) n$$

$$= \left(\frac{V + v_0}{V} \right) n$$

$$= \left(\frac{V + \frac{10}{3}}{V} \right) \times 250$$

$$\therefore n_2 - n_1 = \frac{250}{V} \times \frac{20}{3}$$

$$\text{or, } 5 = \frac{250 \times 20}{3 \times V}$$

$$\text{or, } V = \frac{250 \times 20}{3 \times 5}$$

$$= 333.33 \text{ m/sec.}$$

5 Doppler effect in light

There is a change in the frequency of light radiation when the source or the observer move with respect to one another. This phenomenon is known as Doppler effect and is similar to the apparent change in the pitch of sound when either the source or the observer is in motion with respect to one another.

There is, however, one difference between the Doppler effects in sound and in light. In deriving the Doppler effect in sound, it was found that the change in frequency for a given velocity v depends on whether it is the source or the observer that is moving with that speed. Such a distinction is possible for sound because there is a medium (the air) relative to which the motion takes place, and so it is not surprising that the motion of the source or the observer relative to the still air can be distinguished. Thus Doppler effect in light is *asymmetric*, the apparent frequency when the source is moving towards a stationary observer is different than that when the observer is moving towards a stationary source. In the first case we have

$$n' = \left(\frac{V}{V - v} \right) n$$

Oscillation

$$\text{or, } a \cos \delta = 0$$

from (i) and (ii),

$$a^2 = (0.37)^2$$

$$\text{or, } a = \pm 0.37 \text{ cm.}$$

$$(c) \quad v = w \sqrt{a^2 - y^2}$$

v is maximum when $y = 0$.

$$\begin{aligned} \therefore v_{\max} &= w \cdot a = a \times 2\pi n \\ &= 0.37 \times 2 \times 3.14 \times 0.25 \\ &= 0.5809 \text{ cm/sec.} \end{aligned}$$

$$\begin{aligned} (d) \quad (accln)_{\max} &= -w^2 \cdot a \\ &= -(2\pi n)^2 \cdot a \\ &= -(2 \times 3.14 \times 0.25)^2 \times 0.37 \\ &= -0.912013 \text{ cm/sec}^2. \end{aligned}$$

Example (1.4.) The displacement of an oscillating particle at an instant t is given by

$$y = a \cos wt + b \sin wt.$$

Show that it is executing a simple harmonic motion.

If $a = 5 \text{ cm}$, $b = 12 \text{ cm}$ and $w = 4 \text{ radian/sec}$, calculate (i) the amplitude, (ii) the time period, (iii) the maximum velocity and (iv) the maximum acceleration of the particle.

Soln.

$$y = a \cos wt + b \sin wt$$

$$\text{or, } \frac{dy}{dt} = -aw \sin wt + bw \cos wt$$

$$\text{or, } \frac{d^2y}{dt^2} = -w^2 \cdot a \cos wt - w^2 b \sin wt$$

$$= -w^2 (a \cos wt + b \sin wt)$$

$$= -w^2 y$$

$$\text{or, } \frac{d^2 y}{dt^2} + w^2 y = 0$$

Hence the motion is simple harmonic.

(i) Let $a = A \sin \alpha$ and $b = A \cos \alpha$.

Then

$$\begin{aligned} y &= A \sin \alpha \cos wt + A \cos \alpha \sin wt \\ &= A \sin (wt + \alpha) \end{aligned}$$

This represents a simple harmonic motion with amplitude A .

$$\therefore A^2 \sin^2 \alpha + A^2 \cos^2 \alpha = a^2 + b^2$$

$$\text{or, } A = \sqrt{a^2 + b^2}$$

$$a = 5 \text{ cm}$$

$$b = 12 \text{ cm}$$

$$\begin{aligned} \therefore A &= \sqrt{5^2 + 12^2} \\ &= 13 \text{ cm} \end{aligned}$$

$$(ii) \quad T = \frac{2\pi}{w} = \frac{2\pi}{4} = \frac{\pi}{2} \text{ sec.}$$

$$\begin{aligned} (iii) \quad v_{\max} &= w.A = 13 \times 4 \\ &= 52 \text{ cm/sec} \end{aligned}$$

$$\begin{aligned} (iv) \quad (a_{\text{ccln}})_{\max} &= -w^2.A = -(4)^2 \times 13 \\ &= -208 \text{ cm/sec}^2. \end{aligned}$$

Example 1.5. The positions of a particle executing simple harmonic motion along the x -axis are $x = A$ and $x = B$ at time t and $2t$ respectively. Show that its period of oscillation is given by

$$\therefore 6 = w \sqrt{a^2 - 64} \quad (i)$$

Again $v = 8 \text{ cm/sec}$ when $y = 6 \text{ cm}$

$$\therefore 8 = w \sqrt{a^2 - 64} \quad (ii)$$

Dividing (ii) by (i) and squaring

$$\frac{64}{36} = \frac{a^2 - 36}{a^2 - 64}$$

or, $a = 10 \text{ cm}$.

Substituting $a = 10 \text{ cm}$ in eqn. (i)

$$\therefore 6 = w \sqrt{100 - 64}$$

or, $w = 1 \text{ rad/sec}$.

Hence frequency,

$$n = \frac{w}{2\pi} = \frac{1}{2\pi} \text{ Hz.}$$

time period

$$T = \frac{1}{n} = 2\pi \text{ seconds.}$$

Example (1.7.) A simple harmonic motion is represented by

$$y = 10 \sin \left(10t - \frac{\pi}{6} \right)$$

where y is measured in metres, t in seconds and the phase angle in radians. Calculate (i) the frequency, (ii) the time period, (iii) the maximum displacement, (iv) the maximum velocity and (v) the maximum acceleration and (vi) displacement, velocity and acceleration at time $t = 0$ and $t = 1$ second.

Soln.

$$\text{Here } y = 10 \sin \left(10t - \frac{\pi}{6} \right) \quad (1)$$

Comparing with the displacement equation

$$y = a \sin (wt + \delta) \quad (2)$$

we get, (i) $w = 2\pi n = 10$

or $n = \frac{10}{2\pi} = 1.6 \text{ Hz.}$

(ii) time period, $T = \frac{1}{n} = \frac{2\pi}{10} = 0.63 \text{ sec.}$

(iii) maximum displacement (amplitude)
 $a = 10 \text{ m.}$

(iv) maximum velocity,

$$v_{\max} = w.a = 10 \times 10 = 100 \text{ m/sec.}$$

(v) $(\text{accln.})_{\max} = -w^2 a = -(10)^2 \times 10$
 $= -1000 \text{ m/sec}^2.$

minus sign shows that the acceleration is directed towards the mean position.

(vi) From eqn. (1)

(a) at $t = 0$

$$y = 10 \sin\left(-\frac{\pi}{6}\right) = -5 \text{ m.}$$

velocity, $\frac{dy}{dt} = a w \cos \delta$

$$= 10 \times 10 \cos\left(-\frac{\pi}{6}\right)$$

$$= 100 \times 0.866 = 86.6 \text{ m/sec.}$$

Acceleration, $\frac{d^2y}{dt^2} = -aw^2 \sin \delta$

$$= -10 \times 10^2 \times \sin\left(-\frac{\pi}{6}\right)$$

$$= -10 \times 100 \times 0.5$$

$$= -500 \text{ m/sec}^2.$$

(b) From eqn. (1), at $t = 1$,

displacement

$$y = 10 \sin\left(10 - \frac{\pi}{6}\right)$$

$$= 10 \sin \left(\frac{60 - 3.142}{6} \right)$$

$$= 10 \sin \left(\frac{56.858}{6} \right)$$

$$= 10 \sin (3\pi) \text{ approximately}$$

$$= 10 \sin \pi$$

$$= 0$$

$$\text{velocity, } \frac{dy}{dt} = a w \cos \left(10 - \frac{\pi}{6} \right)$$

$$= a w \cos (\pi) \text{ approximately}$$

$$= 10 \times 10 \times (-1)$$

$$= -100 \text{ m/sec.}$$

$$\text{Accn., } \frac{d^2y}{dt^2} = -a w^2 \sin \left(10 - \frac{\pi}{6} \right)$$

$$= -a w^2 \sin (\pi) \text{ approximately.}$$

$$= 0.$$

Example 1.8. A particle performs simple harmonic motion given by the equation

$$y = 20 \sin (wt + \alpha)$$

If the time period is 30 seconds and the particle has a displacement of 10 cm at $t = 0$, find (i) epoch, (ii) the phase angle at $t = 5$ seconds and (iii) the phase difference between two positions of the particle 15 seconds apart.

Soln.

Here

$$y = 20 \sin (wt + \alpha)$$

$$T = 30 \text{ secs.}$$

$$\therefore w = \frac{2\pi}{T} = \frac{2\pi}{30} = \frac{\pi}{15} \text{ rad/sec.}$$

(i) at $t = 0$, $y = 10$ cm.

$$\therefore 10 = 20 \sin \left(\frac{\pi}{15} \times 0 + \alpha \right)$$

$$\text{or, } \sin \alpha = \frac{10}{20} = 0.5$$

$$\text{or, } \alpha = \frac{\pi}{6} \text{ radian.}$$

(ii) at $t = 5$ sec,

the phase angle $= (wt + \alpha)$

$$= \left(\frac{\pi}{15} \times 5 + \frac{\pi}{6} \right)$$

$$= \frac{\pi}{2}$$

(iii) at $t = 0$ the phase angle

$$\theta_1 = \left(\frac{\pi}{15} \times 0 + \frac{\pi}{6} \right) = \frac{\pi}{6} \text{ radian.}$$

at $t = 15$ sec, the phase angle

$$\theta_2 = \left(\frac{\pi}{15} \times 15 + \frac{\pi}{6} \right)$$

$$= \frac{7\pi}{6} \text{ radian.}$$

$\therefore$ the phase difference,

$$\theta_2 - \theta_1 = \frac{7\pi}{6} - \frac{\pi}{6} = \pi \text{ radian.}$$

Example 1.9. A body describing SHM has a maximum acceleration of $8\pi \text{ m/s}^2$ and a maximum speed of 1.6 m/s . Find the period T and the amplitude a .

Soln :

$$a_{\max} = \omega^2 a \quad (\text{ignoring the minus sign})$$

$$= (2\pi n)^2 a = \left(\frac{2\pi}{T} \right)^2 a = \frac{4\pi^2}{T^2} a = 8\pi \text{ m/s}^2.$$

again $v_{\max} = \omega a = \left(\frac{2\pi}{T} \right) a = 1.6 \text{ m/s.}$

$$\frac{a_{\max}}{v_{\max}} = \frac{2\pi}{T} = \frac{8\pi}{1.6}; \quad \text{or, } T = 0.4 \text{ s.}$$

Then $\frac{2\pi a}{T} = \frac{2\pi a}{0.4} = 1.6; \quad \text{or, } a = \frac{0.4 \times 1.6}{2\pi}$
 $= 0.102 \text{ m}$

Example 1.10 A ball moves in a circular path of 0.15 m diameter with a constant angular speed of 20 rev/min. Its shadow performs simple harmonic motion on the wall behind it. Find the acceleration and speed of the shadow (i) at a turning point of the motion (ii) at the equilibrium position, and (iii) at a point 6 cm from the equilibrium position.

Soln :

Amplitude is the radius of the circle.

$$\therefore a = \frac{0.15}{2} \text{ m} = 0.075 \text{ m}$$

The period and frequency of the shadow are the same as that of the circular motion.

So

$$20 \text{ rev/min} \times \frac{20}{60} \text{ rev./sec.}$$

$$\omega = (2\pi) \times \frac{20}{60} \text{ rad/s} = \frac{2\pi}{3} \text{ rad/sec.}$$

From the general formulas for SHM, we have

(i) at the turning point $v = 0$ and $a = a_{\max}$.

$$\therefore a_{\max} = \omega^2 a = \left(\frac{4\pi^2}{9} \right) (0.075) = 0.329 \text{ m/sec}^2.$$

(ii) at the equilibrium position,

$$a = 0 \text{ and } v = v_{\max}$$

$$\therefore v_{\max} = \omega a = \left(\frac{2\pi}{3} \right) (0.075) = 0.157 \text{ m/sec.}$$

(iii) at $y = 0.06 \text{ m (6 cm)}$

$$v = \omega \sqrt{a^2 - y^2} = \frac{2\pi}{3} \sqrt{(0.075)^2 - (0.06)^2}$$

$$= \frac{2\pi}{3} (0.045) = 0.0942 \text{ m/sec.}$$

$$a = \omega^2 y = \left(\frac{4\pi^2}{9} \right) (0.06) = 0.263 \text{ m/sec}^2.$$

1.5 Energy of a body executing S.H.M.

The acceleration of a particle and hence the force acting on the particle executing simple harmonic motion is, as we know, directed towards its mean or equilibrium position, *i.e.*, opposite to the direction in which the displacement y increases. Work is, therefore, done during the displacement of the particle. Hence the particle possesses *potential energy* (U). As the particle also possesses velocity, it possesses *kinetic energy* (K) too. Thus the mechanical energy E of a particle executing simple harmonic motion is partly kinetic and partly potential. *If no non-conservative forces, such as the force of friction act on the particle, the sum of its kinetic energy and potential energy remains constant.* or,

$$E = K + U = \text{constant.}$$

As the displacement increase, the potential energy increases and the kinetic energy decreases and *vice versa*. But the total energy $E (= K + U)$ is conserved.

Let the displacement of a particle executing simple harmonic motion at any instant be y . If the mass of the particle be m and its velocity

$$\begin{aligned}
 &= \frac{m \omega^2 a^2}{4T} \int_0^T 2 \cos^2 (\omega t + \phi) dt \\
 &= \frac{m \omega^2 a^2}{4T} \int_0^T [1 + \cos 2 (\omega t + \phi)] dt \\
 &= \frac{m \omega^2 a^2}{4T} \left[\int_0^T dt + \int_0^T \cos 2 (\omega t + \phi) dt \right]
 \end{aligned}$$

Again, the average value of a sine or cosine function over a complete cycle or a whole time period is zero. Hence

average K.E. of the particle

$$\begin{aligned}
 &= \frac{m \omega^2 a^2}{4T} [t]_0^T \\
 &= \frac{m \omega^2 a^2}{4T} \cdot T \\
 &= \frac{1}{4} m \omega^2 a^2 = \frac{1}{4} k a^2. \quad (1.19)
 \end{aligned}$$

Thus,

average value of P.E. of the particle

= average value of K.E. of the particle

$$= \frac{1}{4} m \omega^2 a^2 = \frac{1}{4} k a^2$$

= half the total energy.

Example 1.11. (i) What is the mechanical (total) energy of the oscillating system of Example 1.1 ?

Soln.

$$\text{Total energy} = \text{P.E.} + \text{K.E.} = \frac{1}{4} k a^2$$

Now $k = 200 \text{ N/m}$ and $a = 0.04 \text{ m}$ [Ex. 1.1 (i) and (iv)]

$$\therefore \text{Total energy} = \frac{1}{2} (200) (0.04)^2 = \mathbf{0.16 \text{ joules}}$$

Also total energy,

$$\frac{1}{4} ka^2 = 2\pi^2 m a^2 n^2$$

$$= 2 (3.14)^2 (2) (0.04)^2 (159)$$

$$= \mathbf{0.16 \text{ joules}}$$

From Ex. 1.1

$$a = 0.04 \text{ m}$$

$$m = 2 \text{ Kg}$$

$$n = 159 \text{ Hz}$$

(ii) Compute the velocity, the acceleration and the kinetic and potential energies of the body when it has moved in half-way from its initial position toward the centre of motion.

$$v = \pm w \sqrt{a^2 - y^2}$$

At half-way,

$$= \pm (10 \text{ sec}^{-1}) \sqrt{(0.04\text{m})^2 - (0.02\text{m})^2}$$

$$y = \frac{a}{2} = \frac{0.04}{2} \text{ m} = 0.02\text{m}$$

$$= \pm \frac{2\sqrt{3}}{10} \text{ m/sec}^{-1}$$

$$w = 10 \text{ sec}^{-1}$$

$$= \pm \mathbf{0.346 \text{ m sec}^{-1}}$$

$$\text{acceleration} = -w^2 y = - (10 \text{ sec}^{-1})^2 \cdot (0.02 \text{ m})$$

$$= \mathbf{-2.0 \text{ m sec}^{-2}}$$

$$\text{K. E.} = \frac{1}{2} mv^2 = \frac{1}{2} (2) (0.346)^2 \simeq \mathbf{0.12 \text{ joules}}$$

$$\text{P.E.} = \frac{1}{2} ky^2 = \frac{1}{2} (200) (0.02)^2 \simeq \mathbf{0.04 \text{ joules}}$$

$$\therefore \text{Total energy} = \text{P.E.} + \text{K.E.} = (0.04 + 0.12) \text{ joules} \\ = \mathbf{0.16 \text{ joules.}}$$

Note that total energy is constant.

Example 1.12. A block whose mass is 680 gm is at rest on a table top and is fastened to an anchored horizontal spring. The

Thus when the oscillator is half-way to its end point, 25% of the energy is potential and 75% is Kinetic.

Example 1.13. A 0.2 kg mass suspended from a spring describes a simple harmonic motion with a period T of 3 s and amplitude R of 10 cm. At $t = 0$ the mass passed upward through the equilibrium position. (i) Find the force constant k of the spring (ii) Find the displacement, velocity and acceleration of the mass when $t = 1$ s. (iii) Also show that at $t = 1$, the sum of the potential and kinetic energy is equal to $\frac{1}{2} ka$?

Soln.

$$(i) \quad w = \sqrt{\frac{k}{m}} \quad \text{or, } w^2 = \frac{k}{m}$$

$$\therefore (2\pi n)^2 = \frac{k}{m} ; \left(\frac{2\pi}{T}\right)^2 = \frac{k}{m}$$

$$\frac{4\pi^2}{T^2} = \frac{k}{m}; \quad \text{or, } k = \frac{4\pi^2}{T^2} \times m = \frac{4\pi^2}{3^2} \times (0.2) \\ = 0.88 \text{ N/m.}$$

(ii) Let us consider the upward direction as the positive direction. Then from

$$y = a \sin wt, \quad \text{we have}$$

$$\text{at } t = 0, y = 0 \text{ and } v > 0.$$

At $t = 1$ s,

$$y = 0.10 \sin \left[\left(\frac{2\pi}{3} \right) (1) \right] \quad w = \frac{2\pi}{T} \\ = 0.10 \sin 120^\circ \\ = 0.0866 \text{ m.}$$

$$v = \frac{dy}{dt} = wa \cos wt$$

$$= \left(\frac{2\pi}{T} \right) (a) \cos \left[\left(\frac{2\pi}{T} \right) (1) \right]$$

$$= - \left(\frac{2\pi}{3} \right) (0.10) \cos 120^\circ$$

$$= - 0.105 \text{ m/s}.$$

$$\text{accel.} = \frac{d^2 y}{dt^2} = - w^2 a \sin w t$$

$$= - \left(\frac{2\pi}{T} \right)^2 (a) \sin \left[\left(\frac{2\pi}{T} \right) (t) \right]$$

$$= - \left(\frac{2\pi}{3} \right)^2 (0.10) \sin 120^\circ$$

$$= - 0.38 \text{ m/s}^2.$$

$$(iii) E = P. E. + K. E.$$

$$\frac{1}{2} k a^2 = \frac{1}{2} k y^2 + \frac{1}{2} m v^2.$$

$$\left(\frac{1}{2} \right) (0.88) (0.1)^2 = \left(\frac{1}{2} \right) (0.88) (0.0866)^2 + \left(\frac{1}{2} \right) (0.2) (0.105)^2$$

$$4.4 \times 10^{-3} \text{ J} = (3.3 \times 10^{-3} + 1.1 \times 10^{-3}) \text{ J}$$

$$= 4.4 \times 10^{-3} \text{ J}.$$

1.7 Some examples of simple harmonic motion

Some important examples of simple harmonic motion will be examined below.

(i) Motion of a body suspended from a coil spring

Fig. 1.5 shows a coil spring whose upper end is fixed to a rigid support. A mass m is attached to its free end. Let l be the no-load length of the spring as shown in (a). When the load m is attached to the spring, it hangs in equilibrium with the spring extended by an amount Δl as in (b). Under this condition the upward force F exerted

$$\text{or, } \frac{d^2\theta}{dt^2} = -\left(\frac{\kappa}{I}\right)\theta$$

$$\text{or, } \frac{d^2\theta}{dt^2} + \left(\frac{\kappa}{I}\right)\theta = 0 \quad (1.28)$$

Eqn. 1.28 is similar to the differential equation of simple harmonic motion (eqn. 1.2). Hence the angular oscillations of the torsion pendulum is simple harmonic. We can simply substitute angular displacement θ for linear displacement y , rotational inertia for mass m , and torsional constant κ for force constant k . Making these substitutions, it can be shown that the solution of eqn. 1.28 is a simple harmonic oscillation in the angle coordinate θ and is given by

$$\theta = \theta_m \sin(\omega t + \phi) \quad (1.29)$$

Here θ_m is the maximum angular displacement, that is, the amplitude of the angular oscillation. ω in eqn. 1.29 means angular frequency not angular velocity. In eqn. 1.29 $\omega \neq d\theta/dt$.

In Fig. 1.8 the disk oscillates about the equilibrium position $\theta = 0$, the total angular range being $2\theta_m$ (from OQ to OR). By analogy with eqn. 1.12, the period of oscillation is given by

$$T = 2\pi\sqrt{\frac{I}{\kappa}}$$

If κ is known and T is measured, the rotational inertia I about the axis of rotation of any oscillating rigid body can be determined. If I is known and T is measured, the torsional constant κ of any sample of wire can be determined.

Example 1.14. A particle performs simple harmonic motion given by the equation.

$$y = 20 \sin(\omega t + \alpha)$$

If the time period is 30 seconds and the particle has a displacement of 10 cm at $t = 0$, find (i) epoch, (ii) the phase angle at $t = 5$ seconds and (iii) the phase difference between two positions of the particle 15 seconds apart.

Soln.

Here

$$y = 20 \sin (wt + \alpha)$$

$$T = 30 \text{ secs.}$$

$$\therefore w = \frac{2\pi}{T} = \frac{2\pi}{30} = \frac{\pi}{15} \text{ rad/sec.}$$

(i) at $t = 0$, $y = 10 \text{ cm.}$

$$\therefore 10 = 20 \sin \left(\frac{\pi}{15} \times 0 + \alpha \right)$$

$$\text{or, } \sin \alpha = \frac{10}{20} = 0.5$$

$$\text{or, } \alpha = \frac{\pi}{6} \text{ radian.}$$

(ii) at $t = 5 \text{ sec,}$

the phase angle $= (wt + \alpha)$

$$= \left(\frac{\pi}{15} \times 5 + \frac{\pi}{6} \right)$$

$$= \frac{\pi}{2}$$

(iii) at $t = 0$ the phase angle

$$\theta_1 = \left(\frac{\pi}{15} \times 0 + \frac{\pi}{6} \right) = \frac{\pi}{6} \text{ radian.}$$

at $t = 15 \text{ sec,}$ the phase angle

$$\theta_2 = \left(\frac{\pi}{15} \times 15 + \frac{\pi}{6} \right)$$

$$= \frac{7\pi}{6} \text{ radian.}$$

$\therefore$ the phase difference,

$$\theta_2 - \theta_1 = \frac{7\pi}{6} - \frac{\pi}{6} = \pi \text{ radian.}$$

Interference

Thus the expression for the path difference, $\Delta = S_2P - S_1P$, reduces to

$$\Delta = S_2P - S_1P = [D^2 + (x_n + \frac{d}{2})^2]^{1/2} - [D^2 + (x_n - \frac{d}{2})^2]^{1/2}$$

or, $\Delta + [D^2 + (x_n - \frac{d}{2})^2]^{1/2} = [D^2 + (x_n + \frac{d}{2})^2]^{1/2}$

which can be rearranged in the form:

$$\frac{x_n^2}{\Delta^2 / 4} - \frac{D^2}{(d^2 - \Delta^2 / 4)} = 1 \quad (27.26)$$

Eqn. (27.26) represents the equation of a hyperbola in the $X - Y$ plane with S_1 and S_2 as the two foci so long as Δ , the path difference is a constant. Thus the loci of the points of constant path difference are *confocal hyperbolae* with S_1 and S_2 as the common foci.

The curves given by eqn. (27.26), when rotated with the line S_1S_2 as the axis, will trace out hyperboloids in space. The shape of the fringe observed will depend upon the position of the screen. If the screen is held parallel to the line S_1S_2 then the fringe will be approximately *straight lines*. If the screen be placed at right angles to the line S_1S_2 , then the fringes will be cross-sections of the hyperboloids on the screen and hence appear *circular* in shape.

Example (27.1) In a Young's double slit experiment, the separation between the sources are 0.18 mm. The fringes are observed on a screen 90 cm away. If with certain monochromatic source of light, the third bright fringe is situated at a distance of 8.1 mm from the central bright fringe, find the wavelength of light.

Soln.

$$x = \frac{8.1}{3} \text{ mm} = 2.7 \text{ mm} = 2.7 \times 10^{-1} \text{ cm}$$

$$d = 0.18 \text{ mm} = 0.18 \times 10^{-1} \text{ cm},$$

$$D = 90 \text{ cm}, \lambda = ?$$

$$X = \frac{\lambda D}{d} \quad \text{or,} \quad \lambda = \frac{Xd}{D}$$

$$= \frac{2.7 \times 10^{-1} \times 0.18 \times 10^{-1}}{90} \text{ cm}$$

$$= 54 \times 10^{-6} \text{ cm} = 5400 \text{ A.U.}$$

Example 27.2 In a double slit experiment using sodium light of wavelength $5.89 \times 10^{-5} \text{ cm}$, an interference pattern is obtained in which 20 equally spaced fringes occupied a distance of 2.0 cm on screen. On replacing the sodium lamp with another monochromatic source but making no other changes, 30 fringes are found to occupy a distance of 2.4 cm, on the screen. What is the wavelength of light from this source?

Soln.

$$X = \frac{\lambda D}{d}$$

Since D and d are constants X is obviously proportional to λ .

In the first case $X_1 \propto \lambda_1$

In the second case $X_2 \propto \lambda_2$

$$\therefore \frac{\lambda_2}{\lambda_1} = \frac{X_2}{X_1}; \quad \text{or,} \quad \lambda_2 = \lambda_1 \cdot \frac{X_2}{X_1}$$

$$\text{Now } X_1 = \frac{2.0}{20} \text{ cm} = 0.1 \text{ cm and}$$

$$X_2 = \frac{2.4}{30} \text{ cm} = .08 \text{ cm}$$

$$\text{Hence } \lambda_2 = \lambda_1 \cdot \frac{.08}{0.1}$$

$$= 5.89 \times 10^{-5} \times 0.8 \text{ cm}$$

$$= 4.712 \times 10^{-5} \text{ cm} = 4712 \text{ A.U.}$$

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Example 27.4 Light from a sodium vapour lamp ($\lambda = 589 \text{ nm}$) forms an interference pattern on a screen 0.8 m from a pair of slits. The bright fringes in the pattern are 0.35 cm apart. What is the slit separation.

Soln.

$$X = \frac{\lambda D}{d}$$

$$\text{or, } d = \frac{\lambda D}{X}$$

$$\begin{aligned} \therefore d &= \frac{(589 \times 10^{-9}) (0.8)}{(0.35 \times 10^{-2})} \text{ m} \\ &= 1.35 \times 10^{-4} \text{ m} = \mathbf{0.135 \text{ mm.}} \end{aligned}$$

Here

$$\lambda = 589 \text{ nm} = 589 \times 10^{-9} \text{ m}$$

$$D = 0.8 \text{ m}$$

$$X = 0.35 \text{ cm} = 0.35 \times 10^{-2} \text{ m}$$

Example 27.5. In a Young's double-slit experiment, the slits are 2 mm apart and are illuminated with a mixture of two wavelengths $\lambda = 750 \text{ nm}$ and $\lambda' = 900 \text{ nm}$. At what distance from the common central bright fringe on a screen 2 m from the slits will a bright fringe from one interference pattern coincide with a bright fringe from the other?

Soln.

Let the m 'th bright fringe from the λ pattern coincide with the m 'th bright fringe from the λ' pattern.

The distance of the m 'th bright fringe from the central bright fringe

$$Y_m = mX = \frac{m\lambda D}{d}$$

Similarly the distance

$$Y_{m'} = m'X' = \frac{m'\lambda'D}{d}$$

The two distances are equal.

$$\text{So } \underline{m\lambda D} = \underline{m'\lambda'D}$$

Diffraction and polarization

Intensity which decreases steadily as θ increases. This is in agreement with Huygens' principle of secondary waves, for the slit has effectively now become the source of secondary waves.

Finally, if $a < \lambda$, then from the original condition $a \sin \theta = m \lambda$, we get $\sin \theta > 1$ for the first minimum ($m = 1$). This is impossible; therefore, the expression for intensity (eqn. 28.4) does not hold when the width of the slit becomes less than one wavelength. The treatment breaks down.

Example 28.1. A parallel beam of light is incident normally on a narrow slit of width 0.2 mm. The Fraunhofer diffraction pattern is observed on a screen, placed at the focal plane of a convex lens of focal length 20 cm. Calculate the distance between the first two minima and the first two maxima on the screen. Assume $\lambda = 5 \times 10^{-5}$ cm and that the lens is placed very close to the slit.

Soln.

In the Fraunhofer diffraction pattern due to a single slit, the angles of diffraction corresponding to zero intensity are given by

$$a \sin \theta = m \lambda, \quad m = 1, 2, 3, \text{ etc.}$$

For the first minima, $m = 1$

$$\text{or, } \sin \theta_1 = \frac{\lambda}{a} = \frac{5 \times 10^{-5}}{2 \times 10^{-2}}$$

$$\text{or, } \theta_1 = 2.5 \times 10^{-3} \text{ radian.}$$

Let the distance of this first minima from the central maxima be x_1 ; since the lens is very close to the slit, then, very approximately,

$$\sin \theta_1 = \frac{x_1}{f} \quad \text{or, } \theta_1 = \frac{x_1}{f} \text{ radians.}$$

Equating the two values of θ_1 , we get

$$\frac{\lambda}{a} = \frac{x_1}{f}; \quad \text{or, } x_1 = \frac{\lambda}{a} \cdot f$$

$$= 2.5 \times 10^{-3} \times 20 \text{ cm.}$$

Similarly, if θ_2 and x_2 are the angles of diffraction for the second minima ($m = 2$) and the distance of the second minima from the central maxima respectively, we have

$$\theta_2 = \frac{2\lambda}{a} = \frac{x_2}{f}; \quad \text{or, } x_2 = 2 \cdot \frac{\lambda}{a} \cdot f$$

$$= 2 \times 2.5 \times 10^{-3} \times 20 \text{ cm.}$$

Hence the distance of separation between the first two minima
 $= (5.0 - 2.5) \times 10^{-3} \times 20 \text{ cm} = 0.05 \text{ cm.}$

The first maxima occur at $\beta = 1.43 \pi (\simeq \frac{3}{2} \pi)$

$$\text{where } \beta = \frac{\pi a \sin \theta}{\lambda}$$

$$\text{or, } a \sin \theta_1 = 1.43 \lambda (\simeq \frac{3\lambda}{2})$$

$$\text{or, } \theta_1 = \frac{\lambda}{a} \times 1.43 \text{ radians.}$$

Again, if x_1 is the distance of this first maxima from the central maxima, then $\theta_1 = \frac{x_1}{f}$

Equating the two values of θ_1

$$x_1 = \frac{\lambda}{a} \times 1.43 \times f.$$

Similarly, the distance x_2 of the second maxima from the central maxima is given by

$$x_2 = \frac{\lambda}{a} \times 2.46 \times f.$$

Hence, the distance of separation between the first two maxima

$$= \frac{\lambda}{a} \times (2.46 - 1.43) \times f$$

$$= 2.5 \times 10^{-3} \times (2.46 - 1.43) \times 20$$

$$= 0.05 \text{ cm.}$$

Example (28.2) A rectangular aperture of width 0.0025 cm illuminated with parallel beam of monochromatic light wavelength $500 \text{ m}\mu$. A converging lens of focal length 50 cm placed close to the aperture collects the light passing through it produces a diffraction pattern on a screen in its focal plane. Calculate the width of the central maxima on the screen.

Soln.

Let x be the distance of the first minima from the central maxima.

Then

$$x = \frac{\lambda}{a} f \text{ (see example 28.1)}$$

where $\lambda = 500 \text{ m}\mu = 5 \times 10^{-5} \text{ cm}$. ($1 \text{ m}\mu = 1 \times 10^{-3} \times 10^{-4} \text{ cm}$)

$a = \text{width of the slit} = 25 \times 10^{-4} \text{ cm}$.

$f = 50 \text{ cm}$.

$$\therefore x = \frac{50 \times 5 \times 10^{-5}}{25 \times 10^{-4}} = 1 \text{ cm}$$

Now x is also the half linear width of the central maxima. Hence the linear width of the central maxima

$$= 2 \times 1 = 2 \text{ cm.}$$

Example (28.3) A lens whose focal length is 40 cm forms a Fraunhofer diffraction pattern of slit $0.3 \text{ m}\mu$ wide. Calculate the distance of the first dark band and of the next bright band from the axis. (wavelength of light used is $5890 \text{ \AA}$)

Soln.

The angles of diffraction corresponding to zero intensity in the Fraunhofer diffraction pattern due to single slit is given by

$$a \sin \theta = m\lambda \text{ where } m = 1, 2, 3, \text{ etc.}$$

For the first dark band $m = 1$,

$$\text{or, } \sin \theta = \frac{\lambda}{a} = \frac{5890 \times 10^{-8}}{0.3 \times 10^{-1}}$$

$$\text{or, } \theta = \frac{1}{3} \times 589 \times 10^{-5} \text{ radian (for small angles } \sin \theta \simeq \theta).$$

If the distance of the first dark band from the axis be x cm, then the angle of diffraction θ , is also expressed by the relation.

$$\theta = x/f = x/40 \text{ radian.}$$

Equating the two values of θ ,

$$x = \frac{1}{3} \times 40 \times 589 \times 10^{-5} \text{ cm} = 0.785 \text{ mm.}$$

The angle of diffraction corresponding to first bright band on either side of the central band is approximately given by

$$a \sin \theta' = (3/2)\lambda$$

$$\text{or, } \theta' = 3\lambda/2a$$

Hence, the distance x' of the first bright band from the axis is

$$\begin{aligned} x' &= \theta' \times f = (3 \times 5890 \times 10^{-8} \times 40) / 2 \times 0.3 \times 10^{-1} \\ &= 1.178 \text{ mm.} \end{aligned}$$

Example (28.4) Find the half angular width of the central bright maximum in the Fraunhofer diffraction pattern due to a slit of width 12×10^{-4} mm when the slit is illuminated by monochromatic light of wavelength 6×10^{-5} cm.

Soln.

In the relation $a \sin \theta = \lambda$, θ is the half angular width of the central maximum.

$$\text{Hence, } \sin \theta = \frac{\lambda}{a} = \frac{6 \times 10^{-5}}{12 \times 10^{-5}} = 0.50$$

$$\text{or, } \theta = 30^\circ$$

Example (28.5) In a Fraunhofer diffraction due to a narrow slit of width 0.02 cm, the screen is placed 2 metre away from the lens used to obtain the pattern. If the first minima lie 5 mm on either side of the central maxima, find the wavelength of light used.

Soln.

The angles of diffraction corresponding to zero intensity are given by

$$a \sin \theta = n\lambda$$

Here $n = 1$, hence $\sin \theta = \frac{\lambda}{a}$

Again, if the screen is far away from the lens,

$\sin \theta = \frac{x}{f}$ where f is the focal length of the lens or the distance between the screen and the lens. x is the distance of the first minima from the axis.

Then, $\frac{x}{f} = \frac{\lambda}{a}$

or, $\lambda = \frac{ax}{f} = \frac{0.02 \times 0.5}{200} \text{ cm}$

$$= 5 \times 10^{-5} \text{ cm}$$

$$= 5000 \text{ \AA}.$$

28.3 Fraunhofer diffraction at double slit

We shall now study the Fraunhofer diffraction pattern produced by two parallel slits. The arrangement is similar to that of Young's double slit experiment which helped to demonstrate the wave nature of light. In that experiment it was assumed that the width of the slit was comparable with the wavelength of light and that they were very close together. The assumption was essential so that the central maximum is at the centre.

the analyzer is gradually rotated, the intensity of the transmitted light goes on decreasing until it becomes zero, when the optic axes of the two crystals are at right angles to each other. If the rotation of the analyzer is continued then light gradually re-appears until the original intensity of the emergent light is restored when the analyzer has turned through 180° with respect to the polarizer i.e. the two optic axes again becomes parallel to one another. Consider the case when the analyzer has been rotated through an angle θ from its position of maximum intensity. The amplitude A of the plane polarized light incident on the analyzer can be resolved into two components perpendicular to each other. The component $A \cos \theta$ is transmitted while the component $A \sin \theta$ is either rejected or absorbed. Since the intensity is proportional to the square of the amplitude, the intensity I of the transmitted light at this position is given by

$$I \propto A^2 \cos^2 \theta$$

$$\text{Also } I_0 \propto A^2$$

$$\therefore I = I_0 \cos^2 \theta; \quad \text{or, } \frac{I}{I_0} = \cos^2 \theta \quad (30.3)$$

This is known as *Malus' cosine law* according to which the intensity of the polarized light emerging from the analyzer is proportional to the square of the cosine of the angle between the polarizer and the analyzer.

Example 30.1 The refractive index for plastic is 1.25. Calculate the angle of refraction for a ray of light incident at polarizing angle.

Soln.

$$\mu = \tan \phi; \quad \text{or, } 1.25 = \tan \phi$$

$$\text{or, } \phi = \tan^{-1}(1.25) = 51.4^\circ$$

$$\text{Now } \phi + r = 90^\circ$$

$$\text{or, } r = 90^\circ - 51.4^\circ = 38.6^\circ$$

Soln.

From Malus' law $I = I_0 \cos^2 \theta$

$$\text{or, } \frac{I}{I_0} = \cos^2 \theta$$

$$(i) \cos^2 \theta = I/I_0 = 0.125$$

$$\text{or, } \cos \theta = \sqrt{0.125} = 0.3535$$

$$\therefore \theta = \cos^{-1} (0.3535) \simeq 69^\circ.18'$$

$$(ii) \cos^2 \theta = I/I_0 = 0.25$$

$$\text{or, } \cos \theta = \sqrt{0.25} = 0.5$$

$$\therefore \theta = \cos^{-1} (0.5) \simeq 60^\circ$$

$$(iii) \cos^2 \theta = I/I_0 = 0.5$$

$$\text{or, } \cos \theta = \sqrt{0.5} = 0.707$$

$$\therefore \theta = \cos^{-1} (0.707) \simeq 45^\circ$$

$$(iv) \cos^2 \theta = I/I_0 = 0.75$$

$$\text{or, } \cos \theta = \sqrt{0.75} = 0.8666$$

$$\therefore \theta = \cos^{-1} (0.8666) \simeq 30^\circ$$

Example 30.3. The critical angle for certain wavelength of light in the case of a piece of glass is 40° . Find the polarizing angle for glass.

Soln.

Critical angle $C = 40^\circ$.

$$\begin{aligned} \text{Refractive index for glass } \mu &= \frac{1}{\sin C} = \frac{1}{\sin 40^\circ} = \frac{1}{0.6427} \\ &= 1.556 \end{aligned}$$

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From Brewster's law,

$$\mu = \tan i$$

$$\text{or, } i = \tan^{-1} \mu = \tan^{-1} (1.556) = 57.3^\circ$$

Example 30.4. Intensity of light through a polarizer analyzer is maximum when their principal planes are parallel. Through what angle the analyzing nicol must be rotated so that intensity gets reduced to $1/4$ of the maximum value.

Soln.

From Malus' law

$$I_1 = I \cos^2 \theta$$

$$I_1 = \frac{1}{4} = I \cos^2 \theta$$

$$\text{or, } \cos^2 \theta = \frac{1}{4}$$

$$\cos \theta = \pm \frac{1}{2}$$

$$\text{or, } \theta = 60^\circ \text{ or } 120^\circ$$

30.6 Optic axis of a crystal

Let us consider a calcite crystal which is hydrated calcium carbonate (CaCO_3). It was found in large quantities in times gone by in Iceland and is also known as *Iceland spar*. It occurs in nature in great variety of crystal forms (in the rhombohedral class of the hexagonal system), but breaks readily into simple cleavage rhombohedrons or rhombs. Each of the six faces of a rhombohedron is a rhombus.